LITTLE LEMON

Database Management & Sales Analytics Project

Project report based on the supplied final-project brief and LittleLemon_Data.xlsx

1. Project scope

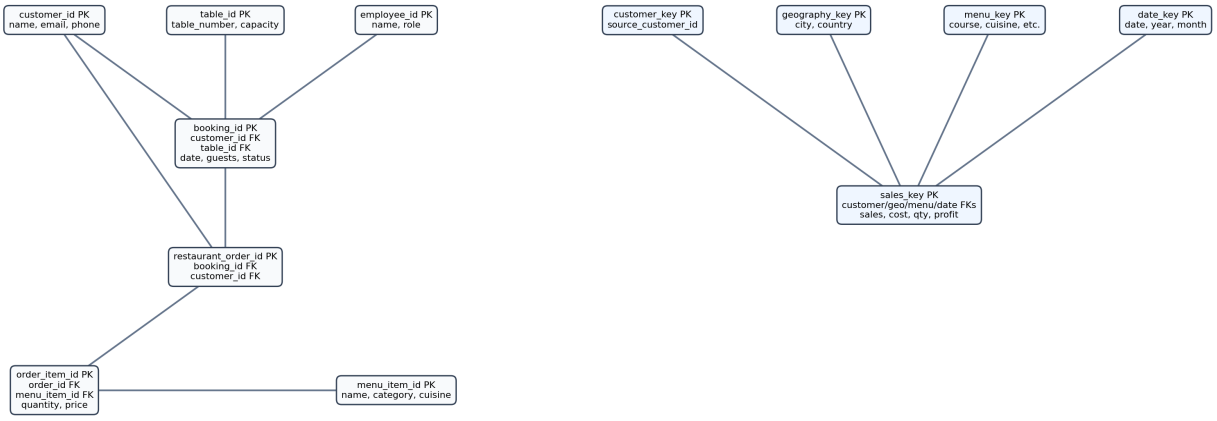
The solution separates the operational restaurant booking system from the analytical sales dataset. The brief requires a database, Python connectivity, procedures that respond to data changes, Tableau connectivity and reporting. The provided spreadsheet contains order-line sales records rather than booking records, so it is loaded into an analytics star schema while bookings are managed in normalized operational tables.

2. Data model

Operational entities: Customers, Employees, RestaurantTables, Bookings, MenuItems, RestaurantOrders and OrderItems. The booking entity links a customer, table and optional employee. Order headers link to customers and optionally to bookings; order items link order headers to menu items.

Analytical entities: DimCustomer, DimGeography, DimMenu and DimDate surround FactSales. FactSales stores one record per source spreadsheet row, with sales, cost, quantity, discount, delivery cost and derived gross profit.

Little Lemon — Operational Relational Model and Analytics Star Schema



3. Dataset profile

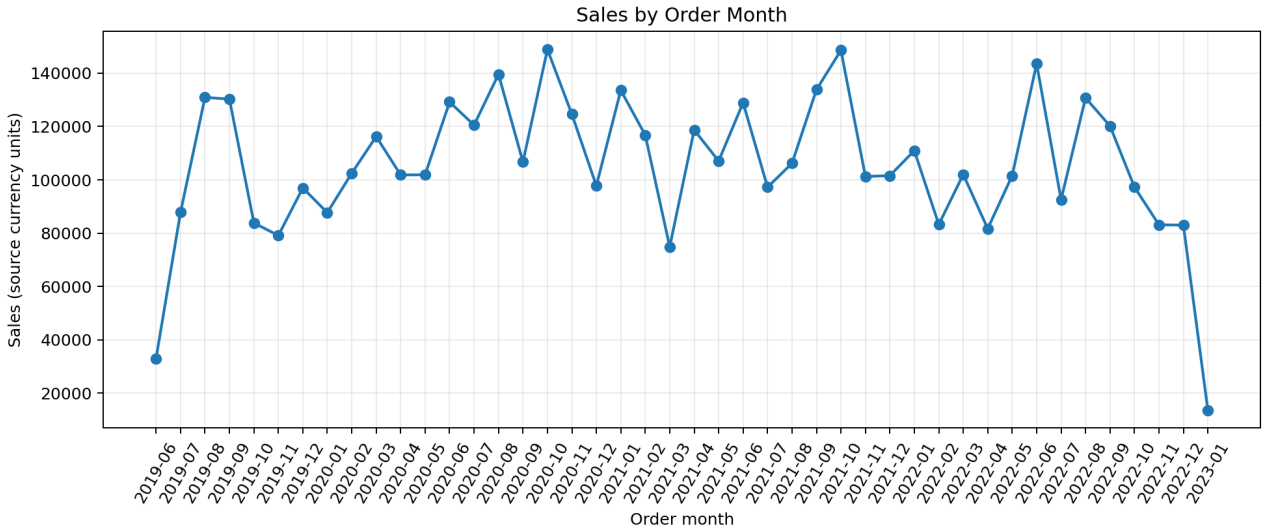
Metric	Observed value
Source rows	21,000
Distinct order IDs	1,000
Distinct customers	1,000
Countries	123
Order date range	2019-06-14 to 2023-01-06
Sales total	4,631,847.03
Cost total	3,087,898.02
Delivery cost total	1,070,484.66
Derived gross profit	473,464.35
Missing country codes	42

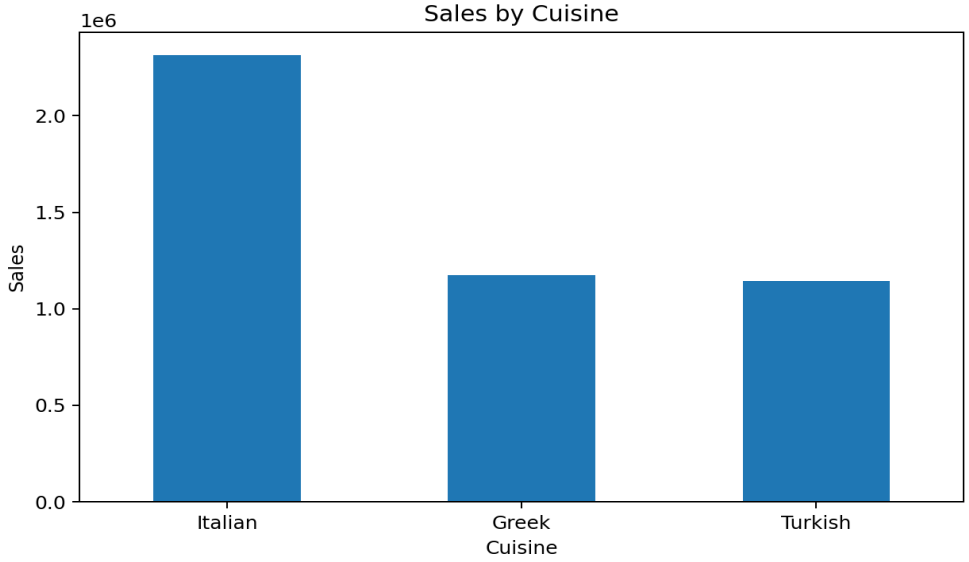
Metric	Observed value
Delivery before order	9,366 rows

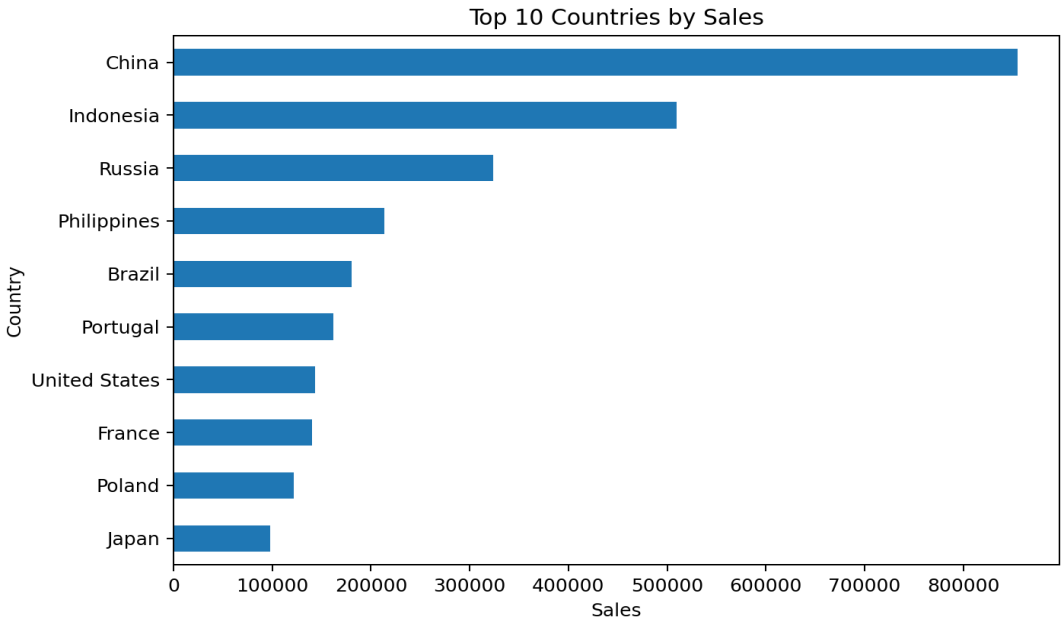
The source workbook does not specify a currency. Financial values are therefore reported in source currency units. Gross Profit is calculated as Sales – Cost – Delivery Cost. The Discount field is preserved as supplied; it is not subtracted a second time because the source does not define whether Sales is pre- or post-discount.

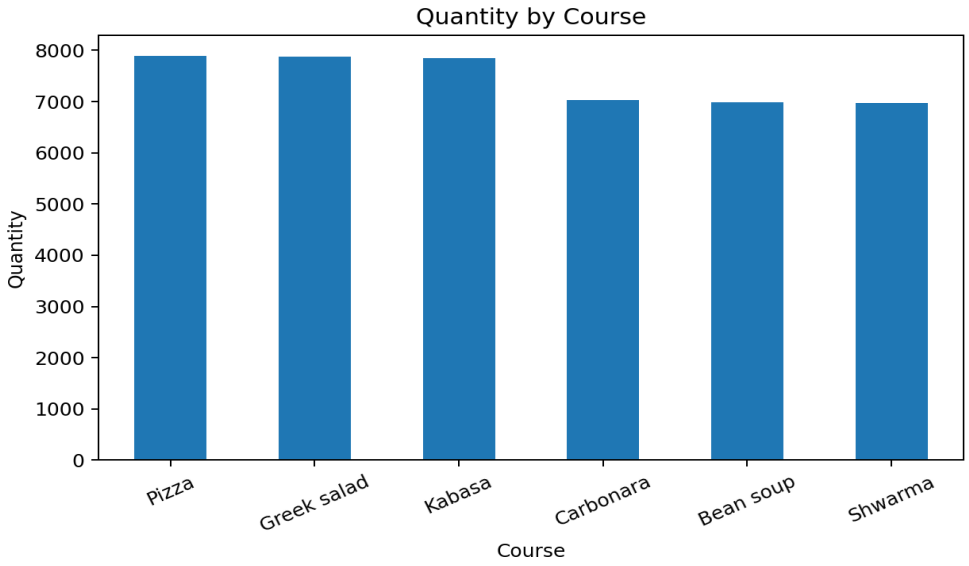
4. Exploratory analysis

Sales by order month, cuisine and country and quantity by course are provided as visual outputs. These views support trend monitoring, product-mix comparisons and geographic concentration analysis. They are descriptive summaries of the provided data, not causal explanations.









5. Data quality findings

There are 42 missing Country Code values; the cleaned analysis extract fills these with UNK without changing the original workbook. A larger issue is that 9,366 rows have Delivery Date earlier than Order Date. Those dates have been retained rather than silently corrected. Validate the source before using delivery lead-time metrics. The workbook contains 21,000 rows but only 1,000 distinct Order IDs, indicating multiple line records per order.

6. Implementation and deployment

Execute sql/little_lemon.sql in MySQL Workbench to create the operational and analytical schemas and the five required procedures. The Python booking client calls those procedures. The ETL script loads the spreadsheet into the dimensional tables and FactSales. Tableau can connect either to the cleaned CSV or directly to the MySQL star schema. See README.md and tableau/TABLEAU_BUILD_GUIDE.md for exact setup steps.

7. Required stored procedures

GetMaxQuantity() returns the largest guest count in the bookings table. ManageBooking(booking_id) checks existence and status. AddBooking(...) validates customer, table capacity and a two-hour table conflict window before inserting. UpdateBooking(...) updates a confirmed booking. CancelBooking(booking_id) changes a confirmed booking to CANCELLED.

8. Limitations and assumptions

The provided brief does not specify table names, booking attributes, restaurant business rules or currency; the operational model and two-hour conflict window are explicit implementation assumptions. The supplied workbook is an order/sales dataset, not booking data. The Tableau workbook is a starter file; complete and save the final packaged workbook in Tableau Desktop after relinking the CSV or configuring the MySQL connection.